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Face presentation attack detection via content-adaptive spatial operators

Shujaat Khan

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Abstract

Face presentation attack detection (FacePAD) is critical for securing facial authentication against print, replay, and mask-based spoofing. This paper proposes CASO-PAD, an RGB-only, single-frame model that enhances MobileNetV3 with content-adaptive spatial operators (involution) to better capture localized spoof cues. Unlike spatially shared convolution kernels, the integrated grouped involution operator generates location-specific kernels conditioned on the input and shares each kernel within a channel group, improving spatial selectivity with limited computational overhead. CASO-PAD remains lightweight (3.6M parameters; 0.256 GFLOPs at $\mathbf{256 \times 256}$) and is trained end-to-end using a standard binary cross-entropy objective. Extensive experiments on Replay-Attack, Replay-Mobile, ROSE-Youtu, and OULU-NPU demonstrate strong performance, achieving 100/100/98.9/99.7% test accuracy, AUC of 1.00/1.00/0.9995/0.9999, and HTER of 0.00/0.00/0.82/0.44%, respectively. On the large-scale SiW-Mv2 Protocol-1 benchmark, CASO-PAD attains 95.45% accuracy with 3.11% HTER and 3.13% EER, indicating strong performance under the evaluated protocol. Ablation studies show that placing the adaptive operator near the network head and using moderate group sharing yields the best accuracy–efficiency balance. Overall, CASO-PAD provides a lightweight RGB-only design for edge-oriented FacePAD under the evaluated settings.

Introduction

Facial recognition is now integral to mobile payments, surveillance, border control, and personal device authentication. Its appeal lies in being non-intrusive and accurate, powering applications such as smartphone unlocking[1], mobile payments[2], surveillance systems[3], access control[4], and attendance monitoring[5]. Leveraging large datasets and high-performance computing, modern recognition systems have surpassed 99% accuracy[6]. Yet, this very sensitivity to facial data introduces vulnerabilities to spoofing attacks, where adversaries use counterfeit inputs such as printed photos, display replays, or 3D masks[7]. As adoption expands into sensitive domains like finance and healthcare, the need for reliable anti-spoofing solutions has become paramount.

FacePAD systems aim to distinguish genuine (*bonafide*) inputs from spoof attempts, ensuring the security of facial recognition technologies[8]. Applications extend beyond traditional security to e-commerce, mobile device

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authentication, and remote access and the demand for robust and deployment-friendly PAD methods has never been greater[9].

Early approaches to FacePAD relied on handcrafted features such as SIFT[10], SURF[11], and HOG[12], often combined with SVMs or LDA classifiers. Texture- and motion-based cues, including eye blinks and lip motion, also proved useful[13,14]. However, these techniques struggled under variable lighting, backgrounds, and camera qualities, limiting their real-world applicability.

Beyond basic photo or video replays, adversaries now employ sophisticated techniques, including high-fidelity 3D masks, adversarial perturbations, and machine learning-driven attacks[15,16] and the arms race between attackers and FacePAD continues. Although data augmentation and unified frameworks such as UniFAD[17] have been proposed, they often require large-scale datasets and remain limited to specific attack scenarios, especially under challenging white-box conditions. Compounding the difficulty, no benchmark dataset can encompass the diversity of possible attack types[18].

To address data scarcity, generative models such as GANs have been explored for synthesizing spoofing samples[19,20]. Recent works demonstrate that adversarial training with synthetic data can improve PAD robustness[21], though challenges of model scalability and efficiency remain. Deep learning, particularly Convolutional Neural Networks (CNNs), has since transformed FacePAD by capturing subtle, discriminative features. CNN-based models perform well but often their high computational cost hinders real-time use in resource-limited environments such as mobile devices.

To deal with the challenges of FacePAD, we propose a content-adaptive framework for resource-constrained environments. The proposed network utilizes a computationally efficient backbone architecture specifically designed for edge devices[22]. The contribution of this work does not lie in proposing the involution principle itself, which is adopted from Li et al.[23]. Instead, it lies in the FacePAD-specific integration of a grouped involution head into a lightweight MobileNetV3 architecture, together with its systematic optimization, kernel-level interpretation, cross-domain evaluation, and deployment-oriented analysis. The resulting operator generates location- and group-dependent kernels conditioned on the input, thereby improving sensitivity to localized spoofing cues while retaining a compact RGB-only, single-frame design. The contributions of this work are as follows:

- An RGB-only, single-frame FacePAD model, CASO-PAD, that augments MobileNetV3[22] with a content-adaptive grouped involution operator[23] for enhanced detection of localized spoofing cues.
- A detailed audit of the learned adaptive kernels, including their spatial, directional, and frequency characteristics, to improve interpretation of the model's decision process.
- A systematic ablation study examining the contribution of the GI module, backbone architecture, operator placement, group count, kernel size, reduction ratio, and input resolution under consistent experimental settings.
- A comprehensive intra-dataset evaluation on Replay-Attack[24], Replay-Mobile[25], OULU-NPU[26], ROSE-Youtu[27], and SiW-Mv2[28], using standard FacePAD metrics including accuracy, AUC, EER, HTER, APCER, BPCER, and ACER.
- An extended generalization analysis comprising OULU-NPU Protocols I and II and leave-one-dataset-out cross-domain evaluation on CASIA-MFSD[18], Replay-Attack[24], MSU-MFSD[8], and OULU-NPU[26], together with deployment-oriented complexity and latency analysis on edge hardware.

The remainder of this paper is structured as follows: In "Related work" section reviews existing FacePAD methods and their limitations. In "Proposed method" section presents the proposed CASO-PAD architecture and grouped involution operator. In "Experiments and results" section describes the experimental setup, datasets, and quantitative results. In "Kernel audit and Ablation studies" sections present the kernel audit and ablation studies, respectively. Finally, "Limitations and future work" section concludes the study.

Related work

FacePAD has attracted substantial interest, producing a broad taxonomy of methods: (i) *spatial/appearance* techniques that analyze frame-level texture and color cues; (ii) *temporal* approaches that exploit motion and dynamics; and (iii) *multimodal* methods that incorporate auxiliary signals such as Remote photoplethysmography (rPPG), depth, thermal, or infrared. Recent work also explores adversarial robustness and domain generalization. Below, we synthesize these directions and highlight gaps motivating an efficient, RGB-only solution.

Spatial/appearance methods

Classical handcrafted features. Early PAD relied on handcrafted descriptors to capture fine-grained textural differences between bonafide and spoof images. Common choices include Local Binary Patterns (LBP)[29], Histograms of Oriented Gradients (HOG)[12], and SIFT[10]. Määttä et al.[30] used LBP to expose print-induced artifacts, while other works explored color spaces (e.g., HSV, YCbCr) and frequency cues to better separate spoof patterns from genuine skin reflectance[31–33]. Although these methods are effective against simple photo attacks, they degrade under varied illuminations, cameras, and high-quality 3D mask or replay attacks[34].

Deep learning and hybrid models. CNN-based FacePAD superseded handcrafted approaches by learning discriminative features end-to-end. Representative advances include two-stream or attention mechanisms over RGB and illumination-invariant spaces, pixel-level supervision for local spoof cues[15], and feature fusion with learned or autoencoder-derived representations[35]. Additional strands investigate adversarial robustness[36], one-class client-specific modeling[37], domain adaptation/generalization[38,39], and diffusion-aided hybrids[40]. Hybrid methods deliberately combine handcrafted texture with CNNs or integrate classical features to improve cross-dataset generalization[41]. While these designs often boost accuracy, many are computationally expensive and less suitable for real-time deployment on edge devices.

Temporal and motion-based methods

Temporal information helps reveal inconsistencies characteristic of replay and mask attacks. Classic works used optical flow or motion patterns[42] and eye-blink analysis via undirected conditional random fields (CRFs)[13]. With deep learning, CNN-RNN hybrids (e.g., CNN+LSTM) capture spatiotemporal dependencies, while deep dynamic textures and flow-guided models further enhance temporal sensitivity[43]. More recent efforts compress or summarize clip dynamics through temporal sequence sampling and stacked recurrent encoders[44,45]. Despite strong accuracy, these approaches are slow in nature due to high latency and can also be parameter- and compute-intensive, limiting use on mobile hardware.

Multimodal signals: rPPG, depth, and infrared

The rPPG leverages subtle color changes induced by blood flow as a liveness prior. Methods range from correspondence features and noise-aware templates to transformer-based encoders[46–48]. rPPG can separate bonafide from spoofed content and has been applied to 3D mask scenarios[49], but it is sensitive to illumination, motion, and video quality, often requiring longer capture windows[50,51]. Beyond rPPG, multispectral and infrared imaging capture non-visible cues and temperature or material differences[52,53]. These systems are powerful but typically demand specialized hardware, limiting scalability in low-cost deployments.

Summary and motivation

Classical texture and motion pipelines are fast but fragile under domain shifts and sophisticated spoofs[34]. Deep models substantially improve accuracy, yet many require heavy backbones, auxiliary sensors, or complex training

that impedes real-time use[40]. Multimodal signals (rPPG, depth, thermal/IR) enhance robustness but demand controlled capture conditions or specialized hardware[49,52,53]. Temporal encoders strengthen replay/mask detection[43–45], but LSTM/transformer stacks are often parameter-intensive for phones and embedded devices.

These trade-offs motivate *efficient, RGB-only* models that retain strong discriminative power while meeting edge constraints. Lightweight backbones such as MobileNetV2/V3 and ShuffleNet achieve favorable accuracy–efficiency via depthwise separable convolutions and attention; however, their spatially shared kernels can under-capture localized spoof artifacts. To address this, *involution*[23] replaces channel-specific, spatially shared kernels with *spatially specific, channel-shared* kernels produced by a lightweight generator, enabling content-adaptive filtering with low overhead.

The adaptation mechanism used by CASO-PAD differs from deformable convolution and other dynamic filtering approaches. Deformable convolution[54] predicts spatial offsets and therefore modifies the locations from which features are sampled. The GI head instead retains a fixed local sampling grid and predicts location- and group-dependent weights for aggregating the sampled features. Dynamic convolution[55] combines a predefined bank of convolution kernels using input-dependent attention, while dynamic filter networks[56] generate input-conditioned convolutional filters. In comparison, the involution formulation generates spatially specific local kernels that are shared across channels within each group. CASO-PAD further differs from prior lightweight FacePAD architectures in that it studies this operator as a selectively placed MobileNetV3 head and evaluates its placement, grouping, kernel size, reduction ratio, learned filtering behaviour, cross-domain generalization, and edge-device performance under a unified experimental framework.

Two recent methods further clarify the positioning of the proposed CASO-PAD framework. SLIP[57] is a one-class FacePAD method that learns from bona fide training images and employs vision–language pretraining, prompt-guided spoof-cue estimation, and liveness-feature disentanglement to improve cross-domain generalization. In contrast, CASO-PAD follows a supervised two-class learning setting and focuses on improving lightweight RGB feature extraction through group-wise, location-adaptive spatial operators. Because both methods report results under common OULU-NPU and leave-one-dataset-out protocols, their quantitative results are compared in Tables 8 and 9, while acknowledging their different one-class and two-class training paradigms.

M3FAS[58], on the other hand, combines RGB images with reflected acoustic signals acquired using a mobile-device camera, speaker, and microphone, and performs hierarchical cross-modal feature fusion. This differs fundamentally from CASO-PAD, which requires only a single RGB frame and no auxiliary sensing hardware. Consequently, a direct numerical comparison is not presented because the two methods use different input modalities, sensing requirements, datasets, and evaluation protocols. Their comparison is therefore conceptual: M3FAS improves robustness through complementary visual–acoustic sensing, whereas CASO-PAD targets efficient RGB-only deployment through content-adaptive spatial processing.

Proposed method

This section presents the proposed CASO-PAD architecture, the formulation and implementation of the grouped involution operator, and the complete forward computational process.

Proposed network architecture

In CNNs, a convolution layer typically apply a single spatial kernel uniformly across the feature map. This entails two well-known constraints:

1. **Spatially agnostic kernels:** the same filter is reused at every location, limiting sensitivity to local texture, scale, and structure.

2. **Fixed cross-channel coupling:** channel interactions are prescribed by the learned $C' \times C$ kernel and remain independent of spatial context.

Before contrasting alternative operators, we fix notation and the receptive-field convention used throughout. Notation. Let $X \in \mathbb{R}^{H \times W \times C}$ denote the input, $Y \in \mathbb{R}^{H \times W \times C'}$ the output, and let $\Omega = \{(u, v) : |u| \leq \lfloor k/2 \rfloor, |v| \leq \lfloor k/2 \rfloor\}$ be the $k \times k$ receptive field.

With this notation, the conventional convolution layer can be written as:

Standard convolution

$$Y(i, j, c_o) = \sum_{c_i=1}^{C} \sum_{(u,v)\in\Omega} K(c_o, c_i, u, v)\, X(i+u, j+v, c_i), \tag{1}$$

where a single kernel K is shared across all spatial positions (i, j).

While standard convolution mixes all channels everywhere, many efficient backbones restrict cross-channel mixing to subsets to reduce cost. This yields group (and in the extreme, depthwise) convolution:

Group convolution

Partition channels into G disjoint groups of size $S = C/G$. Mixing occurs only within the same group:

$$Y(i, j, c_o) = \sum_{c_i \in \mathcal{G}(c_o)} \sum_{(u,v)\in\Omega} K_g(c_o, c_i, u, v)\, X(i+u, j+v, c_i), \tag{2}$$

where $g = g(c_o) \in \{1, \ldots, G\}$ indexes the group of c_o and $\mathcal{G}(c_o)$ is its input-channel set. Standard convolution is $G=1$; depthwise convolution corresponds to $G=C$.

Both of the above still reuse the *same* spatial kernel at every location, which dilutes sensitivity to position-dependent artifacts typical in FacePAD (e.g., specular highlights or print borders). To address this, we adopt a location-adaptive operator.

Involution (location-adaptive, channel-shared)

Involution[23] replaces a *spatially invariant* kernel with *location-specific* kernels $H(i, j, u, v)$ that are shared across channels:

$$Y(i, j, c) = \sum_{(u,v)\in\Omega} H(i, j, u, v)\, X(i+u, j+v, c), \tag{3}$$

where H is generated on-the-fly from features (e.g., via a compact kernel-generator network).

Involution preserves the low compute of depthwise/grouped designs while allowing the spatial kernel to vary across (i, j), a property we will exploit in our content-adaptive head.

Grouped involution head for FacePAD

Building on the grouped involution formulation of Li et al.[23], we adopt it as a lightweight head for FacePAD.. The proposed operator is *group-wise* and *location-adaptive*. For G groups, where one spatial kernel per group and location is generated; that kernel is *shared across channels* within the group and applied depthwise (no channel mixing in the spatial op):

$$Y(i,j,c) = \sum_{(u,v)\in\Omega} H\big(i,j,g(c),u,v\big)\, X(i+u, j+v, c), \tag{4}$$

where $g(c) \in \{1,\ldots,G\}$ maps channel c to its group. Cross-channel interaction is provided by surrounding 1×1 pointwise layers (squeeze/expand), while the spatial operator itself remains group-shared and content-adaptive.

Remarks.

- **Special cases:** $G{=}C$ reduces to depthwise involution[23] (one kernel per channel); $G{=}1$ reduces to channel-shared involution[23].
- **Efficiency:** the spatial application in (4) scales as $\mathcal{O}(C\,k^2HW)$, while the kernel-generator cost scales with group count G and the reduction ratio in the 1×1 bottleneck; both are lightweight compared to full $C{\times}C'$ convolution.

The proposed GI provides a flexible trade-off between expressivity and efficiency by (i) adapting kernels to *spatial content* and (ii) controlling *channel sharing* via the number of groups G. This is especially beneficial for lightweight backbones (e.g., MobileNetV3) and tasks demanding fine spatial awareness (e.g., face PAD, texture analysis, medical imaging), where directional, content-adaptive filtering improves selectivity without incurring the cost of full $C{\times}C'$ spatial mixing.

Proposed MobV3-GI: content-adaptive spatial operator-based network

The proposed network architecture, illustrated in Fig. 1, uses MobileNetV3-Large[22] as a lightweight feature-extraction backbone. The proposed grouped involution operator is integrated near the end of the backbone, where it adaptively transforms the final spatial feature tensor before global feature aggregation and binary classification.

Let $\mathrm{X} \in \mathbb{R}^{B\times C\times H\times W}$ denote the feature tensor supplied to the GI operator, where B, C, H, and W denote the batch size, number of channels, feature-map height, and feature-map width, respectively. The channels are partitioned into G groups, with $S = C/G$ channels in each group. The value of G is selected such that it exactly divides C.

The grouped involution kernels are generated using a lightweight kernel-generation branch. First, a 1×1 pointwise convolution reduces the channel dimension from C to $C_r = \lfloor C/r \rfloor$, where r is the reduction ratio:

$$\mathrm{Q} = \delta\left(\mathrm{BN}\left(\mathrm{Conv}_{1\times1}^{C\to C_r}(\mathrm{X})\right)\right), \qquad \mathrm{Q} \in \mathbb{R}^{B\times C_r\times H\times W}, \tag{5}$$

Here, $\mathrm{BN}(\cdot)$ denotes batch normalization and $\delta(\cdot)$ represents the nonlinear activation function. A second 1×1 pointwise convolution generates Gk^2 kernel coefficients at every spatial location:

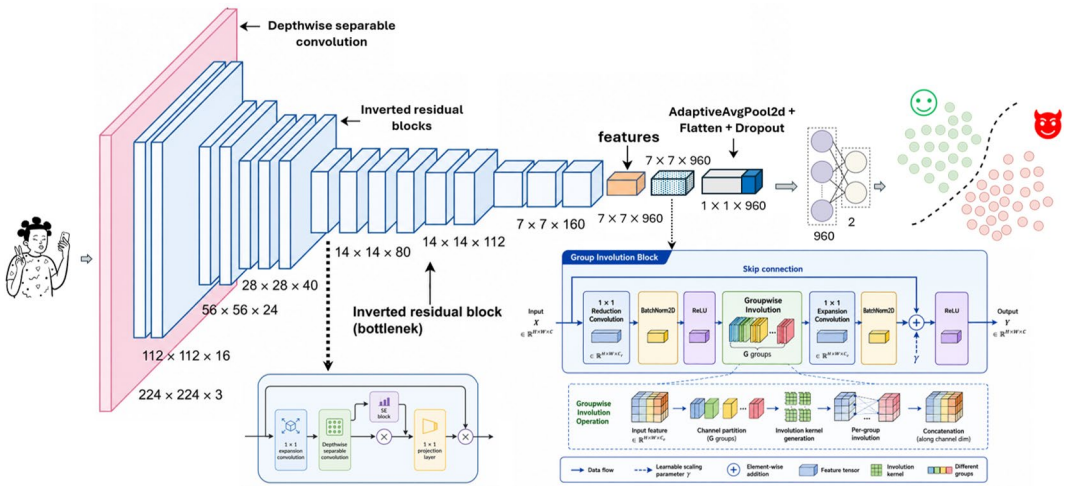


Fig. 1 Schematic diagram of the proposed content adaptive spatial operator-based deep learning network.

$$\mathrm{A} = \mathrm{Conv}_{1\times1}^{C_r \to Gk^2}(\mathrm{Q}), \qquad \mathrm{A} \in \mathbb{R}^{B\times Gk^2\times H\times W}. \tag{6}$$

The generated coefficients are reshaped into group-specific and location-specific spatial kernels as follows:

$$\mathrm{K} = \mathrm{reshape}(\mathrm{A}) \in \mathbb{R}^{B\times G\times 1\times k^2\times H\times W}. \tag{7}$$

The singleton dimension indicates that one generated kernel is shared by all S channels within the corresponding group. Therefore, the kernels vary across spatial locations and channel groups while remaining shared across the channels belonging to the same group.

To perform the feature transformation, same-padded $k \times k$ local neighborhoods are extracted from X using an unfold operation:

$$\mathrm{P} = \mathrm{Unfold}_{k\times k,\, p=\lfloor k/2 \rfloor}(\mathrm{X}) \in \mathbb{R}^{B\times C\times k^2\times H\times W}. \tag{8}$$

The local-patch tensor is subsequently reorganized according to the G channel groups:

$$\mathrm{P}_G = \mathrm{reshape}(\mathrm{P}) \in \mathbb{R}^{B\times G\times S\times k^2\times H\times W}. \tag{9}$$

The generated kernel for each group and spatial location is then applied to the corresponding local feature neighborhood. The output is obtained by summing the weighted responses over the k^2 neighborhood positions:

$$\mathrm{Y}_G(b,g,s,i,j) = \sum_{q=1}^{k^2} \mathrm{K}(b,g,1,q,i,j)\mathrm{P}_G(b,g,s,q,i,j), \tag{10}$$

where $\mathrm{Y}_G \in \mathbb{R}^{B\times G\times S\times H\times W}$. The grouped output is reshaped to restore the original channel organization:

$$\mathrm{Y} = \mathrm{reshape}(\mathrm{Y}_G) \in \mathbb{R}^{B\times C\times H\times W}. \tag{11}$$

Thus, the proposed GI operator preserves the channel and spatial dimensions of its input while replacing a fixed spatial filter with content-dependent kernels that vary across spatial locations and channel groups. Cross-channel interaction is retained through the surrounding 1×1 pointwise transformations in the MobileNetV3 architecture.

Following the grouped involution transformation, global average pooling converts the resulting feature tensor $\mathrm{Y} \in \mathbb{R}^{B\times C\times H\times W}$ into $\mathrm{z} \in \mathbb{R}^{B\times C}$. A linear classification layer with $\mathrm{W} \in \mathbb{R}^{1\times C}$ and $\mathrm{b} \in \mathbb{R}$ generates one binary logit per frame, and the sigmoid function produces the corresponding bona fide probability. The network is optimized using the binary cross-entropy loss described in the Training Setup subsection. Placement. In the default CASO-PAD configuration, the proposed GI operator is placed near the end of the MobileNetV3-Large backbone, immediately before global feature aggregation. This placement allows the content-adaptive operator to process semantically rich features while retaining sufficient spatial information to identify localized spoofing cues. The placement ablation presented in "Ablation studies" section compares this default end-stage placement with an alternative early-stage placement.

Input: A batch of RGB frames $\mathbf{X}_0 \in \mathbb{R}^{B\times 3\times H_0\times W_0}$; number of groups G; kernel size k; and reduction ratio r
Output: Bona fide probabilities $\mathbf{p} \in [0,1]^{B\times 1}$
// Extract the intermediate MobileNetV3 feature tensor
$\mathbf{X} \leftarrow f_{\text{pre}}(\mathbf{X}_0) \in \mathbb{R}^{B\times C\times H\times W}$
$C_r \leftarrow \lfloor C/r \rfloor, \qquad S \leftarrow C/G$
// Generate group- and location-dependent kernels
$\mathbf{Q} \leftarrow \delta\left(\text{BN}\left(\text{Conv}_{1\times 1}^{C\to C_r}(\mathbf{X})\right)\right) \in \mathbb{R}^{B\times C_r\times H\times W}$
$\mathbf{A} \leftarrow \text{Conv}_{1\times 1}^{C_r\to Gk^2}(\mathbf{Q}) \in \mathbb{R}^{B\times Gk^2\times H\times W}$
$\mathbf{K} \leftarrow \text{reshape}(\mathbf{A}) \in \mathbb{R}^{B\times G\times 1\times k^2\times H\times W}$
// Extract and group the local feature neighborhoods
$\mathbf{P} \leftarrow \text{Unfold}_{k\times k,\, p=\lfloor k/2\rfloor}(\mathbf{X}) \in \mathbb{R}^{B\times C\times k^2\times H\times W}$
$\mathbf{P}_G \leftarrow \text{reshape}(\mathbf{P}) \in \mathbb{R}^{B\times G\times S\times k^2\times H\times W}$
// Apply each generated kernel to its corresponding group
$\mathbf{Y}_G \leftarrow \sum_{q=1}^{k^2} \mathbf{K}_{:,:,1,q,:,:} \odot (\mathbf{P}_G)_{:,:,:,q,:,:} \in \mathbb{R}^{B\times G\times S\times H\times W}$
// Restore the original channel organization
$\mathbf{Y} \leftarrow \text{reshape}(\mathbf{Y}_G) \in \mathbb{R}^{B\times C\times H\times W}$
// Perform global feature aggregation and binary classification
$\mathbf{F} \leftarrow \mathbf{Y} \in \mathbb{R}^{B\times C\times H\times W}$
$\mathbf{z} \leftarrow \text{GAP}(\mathbf{F}) \in \mathbb{R}^{B\times C}$
$\mathbf{s} \leftarrow \mathbf{z}\mathbf{W}^{\top} + \mathbf{b} \in \mathbb{R}^{B\times 1}$
$\mathbf{p} \leftarrow \sigma(\mathbf{s}) \in [0,1]^{B\times 1}$
return p

Algorithm 1 Forward computation of CASO-PAD with the proposed grouped involution operator

Algorithm 1 details the complete forward computation of CASO-PAD, with particular emphasis on the internal operations of the proposed GI module. It explicitly describes the kernel-generation branch, extraction of local neighborhoods, channel grouping, application of group- and location-dependent kernels, restoration of the transformed feature tensor, and final binary classification. Tensor dimensions are reported at every major computational stage.

For the default end-placed GI configuration with a 256×256 input, the tensor entering the GI operator has dimensions $\mathrm{X} \in \mathbb{R}^{B\times 960\times 8\times 8}$. Using $G = 120$, $k = 5$, and $r = 4$ gives $S = 8$ channels per group and $C_r = 240$. Accordingly, the reduced feature tensor has dimensions $B\times 240\times 8\times 8$, while the generated coefficient tensor has dimensions $B\times 3000\times 8\times 8$, where $3000 = Gk^2 = 120\times 25$. After reshaping, the generated kernel tensor has dimensions $B\times 120\times 1\times 25\times 8\times 8$.

The extracted local-neighborhood tensor has dimensions $B\times 960\times 25\times 8\times 8$, and its grouped representation has dimensions $B\times 120\times 8\times 25\times 8\times 8$. After element-wise kernel application and summation over the 25 neighborhood positions, the transformed output has dimensions $B\times 120\times 8\times 8\times 8$ in grouped form and is subsequently reshaped to $B\times 960\times 8\times 8$ before global average pooling and binary classification.

Experiments and results

This section describes the datasets, preprocessing procedure, training configuration, and experimental evaluation of CASO-PAD. It then presents the quantitative results and comparisons with existing FacePAD methods.

Datasets

To assess the robustness and generalization capability of the proposed approach, seven well-established face anti-spoofing datasets are used: Replay-Attack[24], Replay-Mobile[25], OULU-NPU[26], ROSE-Youtu[27], SiW-Mv2[28], CASIA-MFSD[18], and MSU-MFSD[8]. The first five datasets are used for intra-dataset evaluation, while CASIA-MFSD and MSU-MFSD are additionally included in the leave-one-dataset-out cross-domain evaluation. A concise summary of their characteristics is provided in Table 1.

Replay-attack

The Replay-Attack dataset[24] comprises 1,300 video sequences encompassing both genuine and attack scenarios, including photos and video replays. The data were collected from 50 individuals under both controlled and variable illumination using different cameras. The dataset is partitioned into disjoint training, development, and testing subsets, ensuring that no subject appears in more than one split.

Replay-mobile

The Replay-Mobile dataset[25] was curated to evaluate mobile-based face recognition and presentation attack detection systems. It consists of 1,190 video samples from 40 participants, captured in diverse lighting settings using smartphone cameras. The attacks include both printed-photo and digital-replay types.

OULU-NPU

The OULU-NPU dataset[26] consists of 4,950 video samples collected from 55 subjects using six mobile devices. The data were acquired across three sessions under varying background scenes and illumination conditions. Presentation attacks were generated using two print and two display devices. The dataset is partitioned subject-disjointly into training and testing sets with 20 subjects each, while the remaining 15 subjects are reserved for validation.

ROSE-Youtu

The ROSE-Youtu dataset[27] includes 3,350 videos from 20 subjects, recorded under multiple illumination conditions and using different camera models. It contains three distinct spoof types: video replay, printed photo, and paper-mask attacks. This dataset poses a considerable challenge for FacePAD research due to its diverse acquisition conditions and attack modalities.

Table 1 Overview of the benchmark datasets and their official evaluation partitions. A dash indicates that the dataset does not provide an official development subset.

Dataset	Year	Participants	Bona Fide / Attack Videos	Train / Dev / Test Partition
Replay-Attack[24]	2012	50	300 / 1,000	15 / 15 / 20 subjects; 360 / 360 / 480 PAD videos
Replay-Mobile[25]	2016	40	550 / 640	12 / 16 / 12 subjects; 312 / 416 / 302 PAD videos
OULU-NPU[26]	2017	55	990 / 3,960	20 / 15 / 20 subjects; 1800 / 1350 / 1800 videos
ROSE-Youtu[27]	2018	20	1,000 / 2,350	10 / – / 10 subjects; 1300 / 350 / 1700 PAD videos
SiW-Mv2[28]	2022	600	785 / 915	Fixed Protocol-I train/test partition files; no official development subset
CASIA-MFSD[18]	2012	50	150 / 450	20 / – / 30 subjects; 240 / – / 360 videos
MSU-MFSD[8]	2015	35	70 / 210	15 / – / 20 subjects; 120 / – / 160 videos

The reported Replay-Attack and Replay-Mobile totals include separate bona fide enrollment subsets, which were excluded from PAD training, threshold selection, and testing. For datasets without an official development subset, a 20% validation subset was held out from the corresponding official training partition for early stopping and threshold selection.

Spoofing in the Wild (SiW-Mv2) Dataset

SiW-Mv2[28] is a large-scale FacePAD benchmark designed to evaluate robustness under a wide variety of spoofing scenarios, comprising 14 different presentation attack categories. The dataset includes 915 spoof videos collected from 600 subjects, along with 785 bona fide recordings from 493 individuals. The attack set covers diverse and challenging conditions such as replay-based attacks, partial manipulation of facial regions, silicone-mask presentations, and paper-based artifacts.

CASIA-MFSD

CASIA-MFSD[18], also referred to as CASIA-FASD, contains 600 videos from 50 subjects. Each subject contributes three bona fide videos and nine attack videos, resulting in 150 bona fide and 450 attack recordings. The attacks include warped photographs, photographs with eye-region cut-outs, and video replays captured at different imaging qualities. The dataset is divided into subject-disjoint training and testing subsets containing 20 and 30 subjects, respectively. In this work, CASIA-MFSD is used as both a source and target domain in the leave-one-dataset-out cross-domain evaluation.

MSU-MFSD

MSU-MFSD[8] contains 280 videos from 35 subjects, including 70 bona fide and 210 attack recordings. The videos were captured using the built-in camera of a MacBook Air and the front-facing camera of a Google Nexus 5. Presentation attacks include printed photographs and video replays displayed using mobile and tablet devices. MSU-MFSD is included as both a source and target domain in the leave-one-dataset-out evaluation to increase the diversity of acquisition devices and presentation instruments.

Pre-processing

In this work, the pre-processing stage involves a center-cropping strategy designed to retain the natural aspect ratio of each frame while avoiding geometric distortion during resizing. For both the training and testing phases, the same procedure is applied to ensure consistency.

The adaptive cropping operation follows the steps below:

- Determine the smaller dimension ρ between the frame height (M) and width (N).
- Extract a centered square region of size $(\rho \times \rho)$ from the original frame.

Figure 2 illustrates the adaptive center-cropping procedure implemented in this study, which serves as the foundation for subsequent data augmentation and normalization.

Since CASO-PAD is designed as a single-frame RGB model, the input videos were converted into frame-level samples through random frame selection. During each training epoch, one frame was randomly selected from each video. Therefore, no fixed temporal interval, such as every k-th frame, or predefined frame-per-second sampling rate was used. This strategy reduces temporal redundancy and computational cost while ensuring that every video contributes equally, regardless of its duration. During testing, one frame was randomly selected from each video for each independent trial. Three independent training and testing trials were performed using different randomly selected frames, and the resulting mean and standard deviation are reported in Table 2.

All dataset partitions were constructed at the subject and video levels before frame extraction. For Replay-Attack and Replay-Mobile, the official training, development, and testing partitions were followed without modification. Thus, frames from the same subject or source video cannot appear in more than one subset. The test sets were not used for training, early stopping, hyperparameter selection, or decision-threshold estimation.

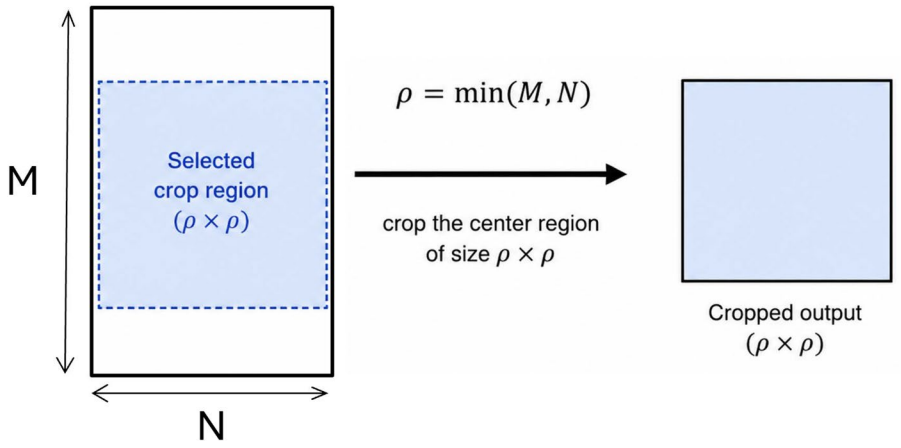


Fig. 2 Illustration of the adaptive center-cropping process used in the pre-processing pipeline. This operation ensures consistent framing and preserves the original aspect ratio across all samples.

Table 2 Performance of the proposed method on Replay-Attack (RA), Replay-Mobile (RM), OULU-NPU, ROSE-Youtu (RY), and SiW-Mv2 (Protocol-1) datasets. Results are reported as mean±std over three runs where applicable.

Metric	RA	RM	OULU	RY	SiW-Mv2
Test Accuracy (%)	100 ± 0.0	100 ± 0.0	99.68 ± 0.13	98.90 ± 0.35	95.45 ± 1.63
Youden Index	1.0 ± 0.0	1.0 ± 0.0	0.991 ± 0.001	0.984 ± 0.004	0.938 ± 0.020
AUC-ROC	1.0 ± 0.0	1.0 ± 0.0	0.9999 ± 0.0000	0.9995 ± 0.0003	0.9906 ± 0.0022
EER (%)	0.0 ± 0.0	0.0 ± 0.0	0.44 ± 0.11	0.82 ± 0.09	3.13 ± 0.70
FAR (%)	0.0 ± 0.0	0.0 ± 0.0	0.23 ± 0.24	0.82 ± 0.09	2.76 ± 0.55
HTER (%)	0.0 ± 0.0	0.0 ± 0.0	0.44 ± 0.04	0.82 ± 0.21	3.11 ± 1.02
FRR (%)	0.0 ± 0.0	0.0 ± 0.0	0.65 ± 0.26	0.82 ± 0.34	3.45 ± 1.75

For each independent trial, the operating threshold was selected by maximizing the Youden index on the official development set where one was provided. For datasets without an official development set, the threshold was selected using the internally held-out validation subset derived exclusively from the training partition. The selected threshold was then fixed for evaluation on the corresponding test set. During testing, exactly one randomly selected frame from each video generated one prediction score. No multi-frame score averaging, majority voting, feature aggregation, or other frame-to-video fusion was applied. Consequently, every test video contributed equally through a single frame-level decision. This evaluation is more restrictive than multi-frame video aggregation, which may reduce prediction errors by combining information across several frames[40,59,60].

Training setup

The training of the proposed model is performed using the Adam optimizer with a learning rate of 10^{-4}. The optimization objective is the binary cross-entropy (BCE) loss, formulated as:

$$L = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)], \tag{12}$$

where y_i denotes the ground-truth label and p_i represents the model's estimated probability for the genuine (live) class.

Training is conducted for 100 epochs using mini-batches of size 32. Early stopping with a patience threshold of 20 epochs is employed to prevent overfitting, based on the validation loss trend. During the testing phase, inference is performed with a batch size of 256 to ensure computational efficiency.

All experiments are implemented in the PyTorch framework and executed on an NVIDIA GeForce RTX 5080 GPU equipped with 16 GB of VRAM. Unless otherwise stated, all input frames are resized to a spatial resolution

of 256×256 pixels. The source code and implementation details are publicly available at: https://github.com/Shujaat123/CASO-PAD.

Performance evaluation

The proposed *Content-Adaptive Spatial Operators for FacePAD* (CASO-PAD) is built on an involution-augmented MobileNetV3 backbone. Table 2 reports the overall performance under the default configuration (256×256 input resolution, G=120 groups, reduction factor r=4, and kernel size k=5). Across five benchmarks, CASO-PAD demonstrates strong discriminative capability, achieving perfect or near-perfect separation on controlled datasets and maintaining robust performance on the more challenging in-the-wild protocol.

On Replay-Attack (RA) and Replay-Mobile (RM), CASO-PAD achieves flawless separation between bona fide and spoof classes, reaching 100% accuracy and AUC-ROC of 1.0 with 0.0% EER/HTER. These results indicate strong generalization under the controlled capture conditions and device variations typical of RA/RM.

On OULU-NPU, which introduces larger variability in illumination, background, capture devices, and attack instruments, CASO-PAD maintains near-ceiling performance (99.68% accuracy, AUC-ROC 0.9999) with low error rates (EER 0.44%, HTER 0.44%).

ROSE-Youtu (RY) is more challenging due to diverse spoof types (print, replay, and mask/paper-mask) and heterogeneous acquisition settings. CASO-PAD sustains strong robustness with 98.90% accuracy, AUC-ROC 0.99, and EER/HTER of 0.82%, while preserving balanced sensitivity and specificity (Youden Index 0.98).

Finally, on SiW-Mv2 Protocol-1, which reflects large-scale, in-the-wild spoofing conditions with substantial subject and attack diversity, CASO-PAD achieves 95.45% accuracy and AUC-ROC 0.9906 with 3.11% HTER and 3.13% EER. Overall, these results confirm that the proposed content-adaptive spatial operator improves discriminability on standard benchmarks while remaining robust under more realistic, unconstrained attack scenarios.

Comparison with the state of the art

To benchmark its performance, we compare against a broad range of contemporary approaches on RA, RM, OULU-NPU, RY, and SiW-Mv2 datasets using standard evaluation metrics.

The comparison tables contain both literature-reported and reimplemented results. Literature-reported results are included only when the corresponding study follows the same official dataset split, evaluation protocol, and metric used in this work. Nevertheless, differences in input resolution, preprocessing, augmentation, and optimization may remain; these entries should therefore be interpreted as protocol-level comparisons. Baselines marked with † were reimplemented and evaluated using the same preprocessing, training, and evaluation pipeline as CASO-PAD.

To emphasize recent developments, the intra-dataset comparison tables primarily include methods published from 2021 onward. A limited number of 2020 methods are retained when they report results under the same official evaluation protocol and remain directly comparable. Earlier methods are retained only in the SiW-Mv2 and cross-domain comparisons for specific reasons. In the SiW-Mv2 comparison, established architectures such as VGG16, InceptionV3, and MobileNetV3 are reimplemented using the same experimental pipeline and provide recognized architectural reference points. In the cross-domain comparison, classical texture-based, auxiliary-supervision, and early domain-generalization methods are retained because they remain standard reference baselines under the widely used C–I–M–O leave-one-dataset-out protocols.

Replay-attack (RA)

As shown in Table 3, CASO-PAD achieves perfect classification on the Replay-Attack dataset with 0.0% HTER, matching or outperforming the most recent high-performing models such as AdvSpoofGuard[21], Dual-Branch[72],

Table 3 Replay-Attack (RA): comparison of test HTER (%) sorted from highest (worst) to lowest (best).

Method	Year	HTER (%)
EfficientNet-B0[61]	2024	36.88
InceptionV4[62]	2020	13.54
3D ConvNet[63]	2023	11.70
SCNN[62]	2020	7.53
Multi-Block LBP[64]	2023	6.98
MIQF+SVM[65]	2022	5.38
GoogLeNet+GMM[37]	2021	3.76
SfSNet[66]	2020	3.10
VGG16+GMM[37]	2021	1.46
MobileNet+Image Diffusion[40]	2023	0.09
ResNet50V2[67]	2023	0.03
HybridNet II[68]	2025	0.02
Lightweight 3D-DNN[69]	2024	0.00
HaTFAS[70]	2024	0.00
HybridNet I[68]	2025	0.00
Deformable Convolution [71]	2025	0.00
Spatio-Temporal [59]	2025	0.00
AdvSpoofGuard [21]	2025	0.00
Dual-Branch [72]	2025	0.00
CASO-PAD (Proposed)	2026	0.00

† Reimplemented and evaluated using the same experimental pipeline as CASO-PAD. All unmarked baseline results are reported from the cited publications under the corresponding official evaluation protocol.

Table 4 Replay-Mobile (RM): comparison of test HTER (%) sorted from highest (worst) to lowest (best).

Method	Year	HTER (%)
VGG16+GMM[37]	2021	17.21
GoogLeNet+GMM[37]	2021	13.56
SMKFNS[73]	2020	11.88
3D ConvNet[63]	2023	8.70
MK-SVDD-Slim[74]	2021	7.60
MKL[74]	2021	6.70
InceptionV4[62]	2020	5.94
WA (PSO+PS)[75]	2021	5.85
WA (GA+MMS+PS)[76]	2022	5.12
SCNN[62]	2020	4.96
EfficientNet-B0[61]	2024	4.62
MobileNet+Image Diffusion[40]	2023	1.14
Lightweight 3D-DNN[69]	2024	0.45
ResNet50V2[67]	2023	0.00
Deformable Convolution [71]	2025	0.00
Spatio-Temporal [59]	2025	0.00
AdvSpoofGuard [21]	2025	0.00
Dual-Branch [72]	2025	0.00
CASO-PAD (Proposed)	2026	0.00

† Reimplemented and evaluated using the same experimental pipeline as CASO-PAD. All unmarked baseline results are reported from the cited publications under the corresponding official evaluation protocol.

and Deformable Convolution[71]. Earlier methods, including EfficientNet-B0[61] and InceptionV4[62], show considerably higher error rates, confirming the progress made by modern lightweight architectures. The zero-error performance achieved by CASO-PAD underscores its strong feature discrimination and adaptation capabilities.

Replay-Mobile (RM)

The results on the Replay-Mobile dataset, summarized in Table 4, follow a similar trend. CASO-PAD again attains 0.0% HTER, positioning it among the top-performing methods such as Dual-Branch[72] and AdvSpoofGuard[21].

Older handcrafted and hybrid approaches (e.g., WA[75,76] and SCNN[62]) show much higher error rates between 5–12%. These findings highlight CASO-PAD's ability to generalize across mobile capture environments characterized by unstable illumination and varying conditions.

OULU

Table 5 reports comparative performance on the OULU-NPU benchmark under the complete protocol, where methods are evaluated using the standardized APCER, BPCER, and ACER metrics. CASO-PAD achieves an ACER of 0.42%, delivering one of the best overall results among recent state-of-the-art approaches. In particular, the proposed method obtains an APCER of 0.00%, indicating perfect rejection of attack presentations, while maintaining a competitive BPCER of 0.83%.

Compared with earlier handcrafted feature baselines such as Texture (VAR)[77] and ED-LBP (VAR)[78], CASO-PAD reduces ACER by more than an order of magnitude. Furthermore, CASO-PAD remains highly competitive against recent deep architectures including 3DLCN[82] (ACER 1.0%) and KD+Depth[72] (ACER 0.56%). These results demonstrate that CASO-PAD generalizes strongly across the diverse illumination, camera, and spoofing conditions present in OULU-NPU.

ROSE-Youtu (RY)

Table 6 reports comparative performance on the more challenging ROSE-Youtu benchmark. Here, CASO-PAD achieves an HTER of 0.82% and an EER of 0.82%, outperforming several recent high-performing methods including Dual-Branch[72] (1.02/1.15) and Deformable Convolution[71] (1.26/0.80). Compared with earlier architectures such as ViViT[83] or EfficientNet-B0[61], the improvement margin exceeds 90%, demonstrating substantial gains in discriminability and generalization. The results on ROSE-Youtu validate CASO-PAD's robustness against diverse spoofing modalities and sensor variations.

Spoofing in the Wild (SiW-Mv2)

Table 7 reports results on the SiW-Mv2 Protocol-1 benchmark, a large-scale and challenging dataset characterized by substantial subject diversity, varied attack instruments, and realistic capture conditions. Under this protocol, CASO-PAD achieves an HTER of 3.11% and an EER of 3.13%, yielding the lowest error rates among the compared methods.

Compared with recent lightweight architectures such as EfficientNet-B0[61] (4.82/5.80) and DeformMobileNet[71] (4.88/4.83), CASO-PAD reduces HTER by approximately 35–40%, indicating improved generalization under diverse spoof scenarios. Relative to ResNet50V2[67] (5.18/6.08) and the depth-augmented teacher model[72]

Table 5 OULU-NPU (complete protocol): comparison of test performance sorted by ACER from highest (worst) to lowest (best).

Method	Year	APCER / BPCER / ACER (%)
Texture (VAR)[77]	2021	14.5 / 15.0 / 14.8
ED-LBP (VAR)[78]	2021	11.3 / 8.4 / 9.9
Fake-Net (VAR)[79]	2021	5.4 / 6.9 / 6.2
OFT (VAR)[80]	2022	5.7 / 2.7 / 4.2
UCDCN[81]	2024	2.6 / 1.01 / 1.82
3DLCN[82]	2024	1.5 / 0.5 / 1.0
Spatio-Temporal [59]	2025	0.13 / 1.11 / 0.62
KD+Depth [72]	2025	0.28 / 0.83 / 0.56
CASO-PAD (Proposed)	2026	0.00 / 0.83 / 0.42

† Reimplemented and evaluated using the same experimental pipeline as CASO-PAD. All unmarked baseline results are reported from the cited publications under the corresponding official evaluation protocol.

Table 6 ROSE-Youtu (RY): comparison of test HTER/EER (%) sorted by HTER from highest (worst) to lowest (best). Entries without reported HTER are listed last.

Method	Year	HTER/EER (%)
3D ConvNet[63]	2023	21.30/-
ResNet50+GMM[37]	2021	14.69/-
ViViT[83]	2023	13.28/2.46
EfficientNet-B0[61]	2024	9.54/-
FASNet[36]	2021	8.57/-
Fatemifar et al.[84]	2022	6.34/-
WA (PSO+PS)[75]	2021	5.61/-
WA (GA+MMS+PS)[76]	2022	5.12/-
MobileNet+Image Diffusion[40]	2023	4.92/4.95
ResNet50V2[67]	2023	2.53/2.64
AdvSpoofGuard [21]	2025	1.97/1.08
Spatio-Temporal (τ=1)[59]	2025	1.47/0.85
CA-FAS[85]	2024	1.37/-
Deformable Convolution [71]	2025	1.26/0.80
Dual-Branch [72]	2025	1.02/1.15
CASO-PAD (Proposed)	2026	0.82/0.82

† Reimplemented and evaluated using the same experimental pipeline as CASO-PAD. All unmarked baseline results are reported from the cited publications under the corresponding official evaluation protocol.

Table 7 SiW-Mv2 (Protocol-1): comparison of test HTER/EER (%) sorted by HTER from highest (worst) to lowest (best).

Method	Year	HTER/EER (%)
VGG16 [86]	2014	10.54/10.45
Spatio-Temporal (τ=1)[59]	2025	6.11/6.63
MobileNetV3 large [22]	2019	5.93/5.87
Depth-aug. Teacher [72]	2025	5.82/6.08
Augmented-Flow (DS) [87]	2026	5.78/6.35
Inceptionv3 [88]	2016	5.77/5.78
Augmented-Flow (I&F) [87]	2026	5.65/6.07
MobileNetV3 small [22]	2019	5.56/5.51
ResNet50V2 [67]	2023	5.18/6.08
DeformMobileNet [71]	2025	4.88/4.83
EfficientNet-B0 [61]	2024	4.82/5.80
CASO-PAD (Proposed)	2026	3.11/3.13

† Reimplemented and evaluated using the same experimental pipeline as CASO-PAD. All unmarked baseline results are reported from the cited publications under the corresponding official evaluation protocol.

(5.82/6.08), the reduction in HTER exceeds 40%, despite relying solely on RGB input without auxiliary depth or temporal modeling.

Furthermore, CASO-PAD substantially outperforms earlier baselines such as MobileNetV3[22] and VGG16[86], where error rates remain above 5–10%. These results demonstrate that the proposed content-adaptive spatial operator enhances discriminative capability in complex, real-world spoofing conditions while preserving a lightweight and deployment-friendly architecture.

Overall discussion

Across the evaluated datasets, CASO-PAD achieves competitive protocol-level results while maintaining a compact and computationally efficient architecture. Because several comparison values are literature-reported, these results should not be interpreted as controlled evidence of universal real-world superiority. The cross-domain experiments further show that performance remains dependent on the unseen target distribution.

Domain generalization analysis

OULU-NPU protocol I and protocol II

To directly address the generalization behavior of CASO-PAD under unseen spoofing conditions, we additionally evaluate the proposed model on OULU-NPU Protocol I and Protocol II. Protocol I evaluates generalization under unseen environmental conditions, whereas Protocol II is particularly relevant to generalization across unseen presentation attack instruments because it evaluates previously unseen print and video-replay attack instruments. Therefore, Protocol II provides a suitable benchmark for assessing whether the model can detect spoofing instruments that are absent from the training set.

It should be emphasized that Protocol II does not hold out an entire attack category: both the training and test sets contain print and video-replay attacks. Instead, it holds out the presentation attack instruments used to generate them. Specifically, training and development use print attacks generated with the Canon imagePRESS C6011 printer and replay attacks displayed on the Dell UltraSharp 1905FP monitor, whereas testing uses the unseen Canon PIXMA iX6550 printer and the early-2015 13-inch MacBook Retina display.

As shown in Table 8, CASO-PAD achieves 2.08% APCER, 6.67% BPCER, and 4.38% ACER on Protocol I. This result indicates moderate robustness to environmental variation, although the relatively higher BPCER suggests that some bona fide samples are falsely rejected under unseen environmental conditions. For additional context, the recent one-class SLIP model reports 12.36% APCER, 16.80% BPCER, and 14.58% ACER under the same protocol. Since SLIP is trained under a one-class setting using only bona fide training samples, this comparison should be interpreted with consideration of the different learning paradigms.

More importantly, on Protocol II, which evaluates unseen print and video-replay attack instruments, CASO-PAD achieves 0.83% APCER, 2.78% BPCER, and 1.81% ACER. The low APCER indicates that only a small fraction of attacks generated using unseen instruments are incorrectly accepted as bona fide. Compared with recent methods from 2022 onward, CASO-PAD achieves lower Protocol II ACER than TSS with ResNet[44], PFTD-MHSA[89], MIQF+SVM[65], M3D[90], MSCI-DSCNN[91], MV+SMCL+TFF[92], AlexNet+PNFG[93], and the reported one-class SLIP result[57]. SLIP obtains 22.16% APCER, 23.18% BPCER, and 22.67% ACER on Protocol II, whereas CASO-PAD obtains 1.81% ACER. These results demonstrate that the proposed MobileNetV3-based content-adaptive operator generalizes effectively to unseen spoofing instruments, while Protocol I suggests that unseen environmental variation remains relatively more challenging.

For comparison, under the standard intra-dataset OULU-NPU evaluation reported in Table 5, CASO-PAD obtains 0.00% APCER, 0.83% BPCER, and 0.42% ACER. Under Protocol II, which holds out the presentation instruments used during testing, these values increase to 0.83%, 2.78%, and 1.81%, respectively. This corresponds to increases of 0.83, 1.95, and 1.39 percentage points in APCER, BPCER, and ACER. The comparison indicates that

Table 8 Performance comparison on OULU-NPU Protocol I and Protocol II.

Method	Year	Protocol I			Protocol II		
		APCER (%)	BPCER (%)	ACER (%)	APCER (%)	BPCER (%)	ACER (%)
TSS with ResNet[44]	2022	0.6	10.3	5.5	2.0	2.1	2.1
PFTD-MHSA[89]	2022	2.4	14.2	8.3	18.3	11.3	14.8
MIQF + SVM[65]	2022	6.9	1.5	4.2	7.8	1.4	4.6
M3D[90]	2023	0.0	1.7	0.8	9.2	0.0	4.6
MSCI-DSCNN[91]	2023	0.3	2.9	1.6	3.1	7.2	5.2
MV + SMCL + TFF[92]	2024	0.5	0.2	0.4	2.9	1.5	2.2
AlexNet + PNFG[93]	2024	0.0	63.2	31.6	5.0	10.4	7.7
SLIP (one-class)[57]	2025	12.36	16.80	14.58	22.16	23.18	22.67
CASO-PAD (Proposed)	2026	2.08	6.67	4.38	0.83	2.78	1.81

† Reimplemented and evaluated using the same experimental pipeline as CASO-PAD. All unmarked baseline results are reported from the cited publications under the corresponding official evaluation protocol.

unseen presentation instruments increase the evaluation difficulty, particularly through a higher rejection rate for bona fide samples. However, this experiment should not be interpreted as evaluation on a completely unseen semantic attack category because print and video-replay attacks occur in both the training and testing partitions.

Cross-dataset domain-generalization evaluation

To further evaluate the domain-generalization capability of CASO-PAD, we conducted leave-one-dataset-out cross-domain experiments using four commonly used FacePAD datasets: CASIA-MFSD (C)[18], Idiap Replay-Attack (I)[24], MSU-MFSD (M)[8], and OULU-NPU (O)[26]. In each setting, three datasets are used for training and the remaining dataset is used exclusively for testing. Therefore, the target dataset is completely unseen during training. The four evaluation protocols are O&M&I→C, O&C&M→I, O&C&I→M, and I&C&M→O.

Table 9 reports the cross-domain comparison. CASO-PAD achieves an average HTER of 17.16% and an average AUC of 88.09% across the four leave-one-dataset-out protocols. The best cross-domain performance is observed on O&C&I→M, where CASO-PAD obtains 6.67% HTER and 98.47% AUC, indicating strong generalization when MSU-MFSD is used as the unseen target domain. In terms of average HTER, CASO-PAD outperforms SLIP, SSDG-M, MADDG, Auxiliary Depth, Binary CNN, MMD-AAE, ColorTexture, LBPTOP, MS-LBP, and IDA. In particular, compared with the reported one-class SLIP results, CASO-PAD reduces the average HTER from 21.89% to 17.16%, while improving the average AUC from 83.70% to 88.09%.

Compared with the related architectural baselines, CASO-PAD reduces the average HTER from 22.77% to 17.16% relative to the Bahdanau Attention model, from 20.23% to 17.16% relative to DeformableMobileNetV2, and from 20.11% to 17.16% relative to the Spatio-Temporal model. These results indicate that the proposed content-adaptive spatial operator improves overall cross-domain robustness beyond the attention-based, deformable, and spatio-temporal variants. Nevertheless, performance varies across individual target domains; for example, DeformableMobileNetV2 and Spatio-Temporal obtain slightly lower HTER on O&C&I→M, while SLIP obtains a lower HTER on O&C&M→I. CASO-PAD, however, provides the lowest average HTER across all four unseen target domains.

Table 9 Leave-one-dataset-out cross-domain comparison on CASIA-MFSD (C), Idiap Replay-Attack (I), MSU-MFSD (M), and OULU-NPU (O). Results are reported as HTER/AUC (%).

Method	Year	I&C&M → O	O&C&I → M	O&C&M → I	O&M&I → C	Avg. HTER/AUC
IDA[8]	2015	54.20/44.59	66.67/27.86	28.35/78.25	55.17/39.05	51.10/47.44
MS-LBP[30]	2011	50.29/49.31	29.76/78.50	50.30/51.64	54.28/44.98	46.16/56.11
LBPTOP[94]	2014	53.15/44.09	36.90/70.80	49.45/49.54	42.60/61.05	45.53/56.37
ColorTexture[31]	2016	63.59/32.71	28.09/78.47	40.40/62.78	30.58/76.89	40.67/62.71
MMD-AAE[95]	2018	40.98/63.08	27.08/83.19	31.58/75.18	44.59/58.29	36.06/69.94
Binary CNN[96]	2014	29.61/77.54	29.25/82.87	34.47/65.88	34.88/71.94	32.05/74.56
Auxiliary Depth[97]	2018	30.17/77.61	22.72/85.88	29.14/71.69	33.52/73.15	28.89/77.08
MADDG[98]	2019	27.98/80.02	17.69/88.06	22.19/84.99	24.50/84.51	23.09/84.40
Bahdanau Attention (τ=3) [60]	2026	36.28/68.20	5.83/97.46	21.00/85.10	27.96/77.70	22.77/82.11
SLIP (one-class)[57]	2025	29.15/77.14	18.81/85.55	15.71/89.38	23.89/82.73	21.89/83.70
SSDG-M[99]	2020	25.17/81.83	16.67/90.47	18.21/94.61	23.11/85.45	20.79/88.09
Deformable MobileNetV2 [71]	2025	23.16/83.53	6.25/96.98	18.00/88.82	33.52/70.30	20.23/84.91
Spatio-Temporal (τ=3) [59]	2025	25.14/81.70	4.58/98.35	25.75/76.94	25.00/80.51	20.11/84.38
CASO-PAD (Proposed)	2026	20.42/87.57	6.67/98.47	19.88/87.04	21.67/79.27	17.16/88.09

† Reimplemented and evaluated using the same experimental pipeline as CASO-PAD. All unmarked baseline results are reported from the cited publications under the corresponding official evaluation protocol.

Kernel audit

To better understand the adaptive behavior of the proposed model, we performed a comprehensive audit of the learned kernel functions. This analysis is conducted on RY dataset and it focuses on four complementary indicators that collectively describe the kernel's spatial and spectral properties:

- **HF/LF ratio** - the proportion of high- to low-frequency energy, where higher values indicate sharper, edge-rich responses;
- **Anisotropy** - the degree of orientation selectivity (0 indicates isotropic kernels, while larger values reflect stronger directional sensitivity);
- **DC offset** - the mean value of kernel weights, where values near zero correspond to center–surround or high-pass behavior without brightness bias; and
- **Position variance** - a measure of spatial non-stationarity, with lower values denoting greater position invariance.

Test-set summary. The evaluation on the held-out test set reveals that the involution-based head produces kernels that are nearly **zero-mean and spatially invariant** (DC offset $\approx 0.0000 \pm 0.0000$, position variance $\approx 4 \times 10^{-6} \pm 3 \times 10^{-6}$). These kernels exhibit **moderate orientation selectivity** (anisotropy 0.1832 ± 0.0696) and a noticeable **high-frequency preference** (HF/LF 17.58 ± 4.65).

Figure 3 illustrates these properties. The mean kernel (left) shows a localized polarity contrast, similar to an oriented edge detector rather than a symmetric center–surround pattern. The corresponding energy map (right) displays a radially concentrated distribution, characteristic of high-pass filtering. Together, these visualizations confirm that the model learns to enhance structural gradients and texture cues typical of genuine facial regions, while naturally suppressing smooth or redundant low-frequency content.

Class-wise differences. A class-wise breakdown of the learned kernels reveals clear distinctions between genuine and spoofed inputs. As depicted in Fig. 4, attack samples exhibit a **higher HF/LF ratio** (18.46 vs. 15.02) but a **lower anisotropy** (0.161 vs. 0.248) than genuine faces. This indicates that spoofed frames tend to contain artificially sharp, broadband textures but lack coherent directional organization, consistent with reflections or printing artifacts.

Cohen's effect size further quantifies these differences: HF/LF ($d = +0.782$) suggests moderately higher sharpness in attacks, while anisotropy ($d = -1.486$) reflects a large effect favoring genuine faces with stronger directional coherence. The histograms in Fig. 4 illustrate these tendencies: real samples cluster around higher anisotropy values, while attack samples dominate the higher HF/LF range. Overall, anisotropy emerges as the more discriminative feature, capturing structural regularity inherent in authentic facial geometry, whereas HF/LF primarily responds to over-sharpening and specular effects common in spoof attempts.

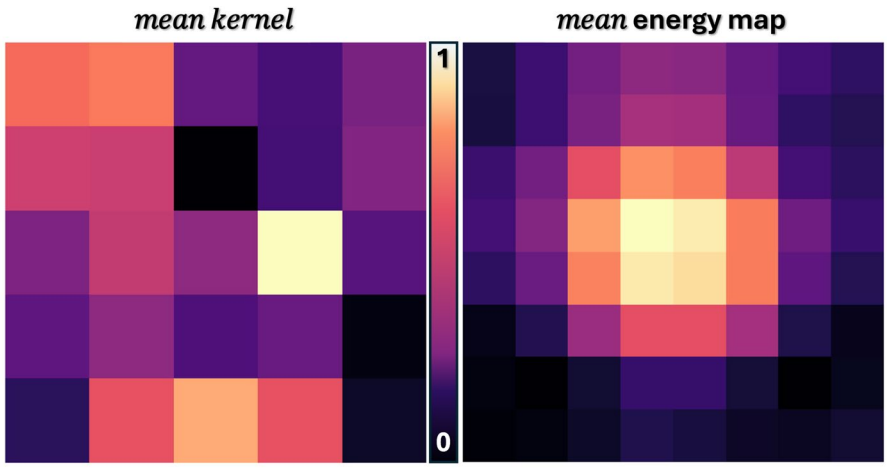
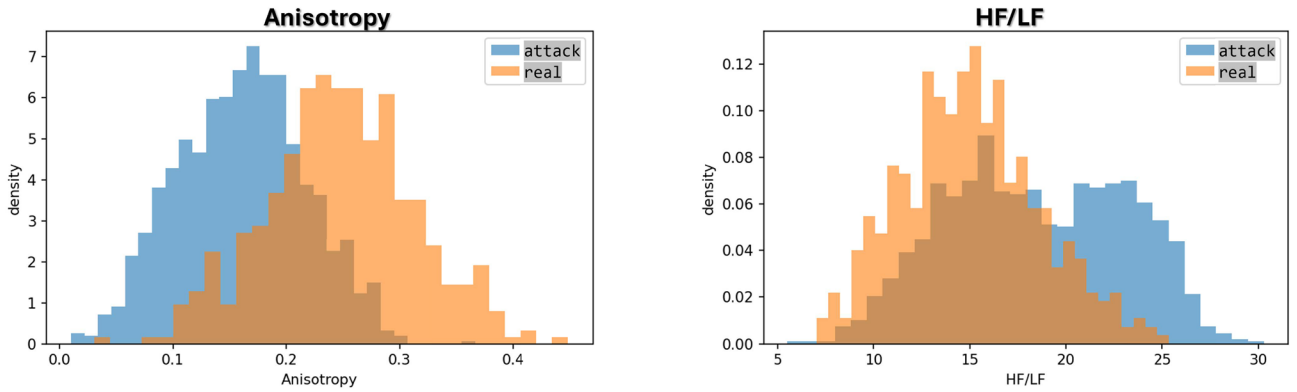


Fig. 3 Kernel audit visualization (normalized to [0, 1]. **Left:** Mean kernel showing directional polarity contrast, resembling an edge-detecting filter. **Right:** Mean energy distribution, radially compact and indicative of high-pass/edge-emphasizing behavior.



(a) Anisotropy distribution **(b)** HF/LF distribution

Fig. 4 Kernel audit overlays on the test set. (**a**) Real faces show higher directional consistency (anisotropy), while (**b**) attack samples exhibit excessive sharpness (higher HF/LF) but weaker organization.

Table 10 Effect of integrating the GI module into MobileNetV3-Large. Results are reported as mean±std HTER/EER (%) over three runs.

Model	ROSE-Youtu	SiW-Mv2 Protocol-1
MobileNetV3-Large	2.48 ± 0.31 / 1.31 ± 0.24	5.93 ± 0.42 / 5.87 ± 0.38
MobileNetV3-Large+GI	0.82 ± 0.21 / 0.82 ± 0.09	3.11 ± 1.02 / 3.13 ± 0.70

Interpretation. The kernel audit suggests that the involution layers act as *adaptive edge-oriented spatial filters*. They enhance structured gradients aligned with genuine facial geometry while down-weighting homogeneous or specular regions often found in spoof media. The combination of near-zero DC bias, balanced frequency response, and moderate directional selectivity highlights the interpretability of the learned filters. In essence, the model implicitly learns a physics-consistent representation, accentuating meaningful texture and geometric cues that distinguish real from fake facial imagery.

Ablation studies

This section presents a series of ablation experiments designed to evaluate the contribution of different architectural and training factors to the proposed method's overall performance. We systematically analyze the effects of network backbone, group count in the involution head, size of kernel, placement of the proposed group-involution (GI) module, input image resolution, and reduction ratio. The section concludes with computational complexity analysis and qualitative interpretation using Grad-CAM visualizations.

Effect of GI module integration

To directly isolate the contribution of the GI module, we compare the original MobileNetV3-Large backbone with the proposed MobileNetV3-Large+GI configuration. Both models were trained and evaluated using identical dataset splits, preprocessing, frame sampling, optimization, and evaluation settings. The comparison was conducted on ROSE-Youtu and SiW-Mv2 Protocol-1, which provide more challenging and less saturated evaluation conditions than Replay-Attack and Replay-Mobile.

As shown in Table 10, integrating the GI module reduces HTER on ROSE-Youtu from 2.48% to 0.82%, an absolute reduction of 1.66 percentage points and a relative reduction of approximately 66.9%. The EER decreases from 1.31% to 0.82%, corresponding to an absolute reduction of 0.49 percentage points and a relative reduction of

approximately 37.4%. On SiW-Mv2 Protocol-1, GI reduces HTER from 5.93% to 3.11% and EER from 5.87% to 3.13%, corresponding to relative reductions of approximately 47.6% and 46.7%, respectively.

A two-sided Welch t-test across the three independent runs shows statistically significant HTER reductions on ROSE-Youtu ($p = 0.0026$) and SiW-Mv2 Protocol-1 ($p = 0.0274$). The EER reduction is significant on SiW-Mv2 Protocol-1 ($p = 0.0087$), while the ROSE-Youtu EER difference does not reach the 0.05 significance level ($p = 0.0573$). These controlled comparisons demonstrate that the observed performance gains are attributable to the content-adaptive GI module rather than solely to the MobileNetV3-Large backbone.

Effect of backbone architecture

To clarify the selection of MobileNetV3-Large and examine how backbone design influences FacePAD performance, we evaluated several established architectures, including VGG16 and VGG19[86], ResNet18 and ResNet50[100], ConvNeXt-Tiny[101], ShuffleNetV2[102], MobileNetV2[103], and MobileNetV3-Large[22]. Each backbone was integrated with the same GI-head design and evaluated on the ROSE-Youtu dataset using the same ImageNet-pretrained initialization strategy[104], identical dataset splits, preprocessing, frame sampling, optimization, and evaluation settings. Because the evaluated backbones produce feature tensors with different channel dimensions, the number of GI groups was selected separately for each backbone such that it exactly divided the corresponding output-channel dimension. The reduction ratio and kernel size were kept fixed at $r = 4$ and $k = 5$, respectively.

As shown in Table 11, the proposed MobileNetV3-Large+GI configuration achieves the lowest HTER and EER, both equal to 0.82%, among the evaluated backbone architectures. ConvNeXt-Tiny+GI provides the closest performance, with an HTER of 0.99% and an EER of 1.00%, followed by ResNet50+GI, which achieves an HTER of 1.25% and an EER of 1.39%. MobileNetV3-Large+GI also outperforms MobileNetV2+GI, reducing HTER from 1.43% to 0.82% and EER from 1.31% to 0.82%.

The results further show that increasing backbone depth or model capacity does not necessarily improve FacePAD performance. In particular, VGG19+GI obtains the highest errors despite its deeper architecture, whereas the lightweight MobileNetV3-Large backbone provides the strongest discrimination when combined with the GI head. This improvement may be attributed to the efficient inverted residual design, squeeze-and-excitation modules, and nonlinear activation functions of MobileNetV3, which complement the spatially adaptive processing performed by the GI operator. Therefore, MobileNetV3-Large was selected as the default backbone based on both its controlled empirical performance and its suitability for resource-constrained deployment.

Effect of group count in the GI head

The number of groups (G) in the involution operator controls spatial diversity and computational load. Table 12 summarizes performance for various group counts. Performance is highest at $G = 120$. Lower and higher group counts yield larger HTER values in these experiments, indicating that both insufficient and excessive kernel diversity can be unfavorable.

Table 11 Comparison of different backbone architectures integrated with the proposed GI head on the ROSE-Youtu dataset ($k = 5$, $r = 4$). Results are reported as mean±std over three runs.

Model	Params (M)	GFLOPs	EER (%) ↓	HTER (%) ↓
VGG16+GI[86]	14.905	15.356	1.92 ± 0.27	2.07 ± 0.31
VGG19+GI[86]	20.215	19.518	4.08 ± 0.36	3.30 ± 0.29
ResNet18+GI[100]	11.367	1.828	1.62 ± 0.24	1.43 ± 0.20
ResNet50+GI[100]	26.506	4.256	1.39 ± 0.18	1.25 ± 0.16
ConvNeXt-Tiny+GI[101]	28.244	4.490	1.00 ± 0.14	0.99 ± 0.12
ShuffleNetV2 ×1.0+GI[102]	2.007	0.184	2.16 ± 0.30	1.79 ± 0.26
MobileNetV2+GI[103]	3.399	0.370	1.31 ± 0.22	1.43 ± 0.34
MobileNetV3-Large+GI[22]	**3.635**	**0.256**	0.82 ± 0.09	0.82 ± 0.21

Table 12 Ablation over the group count (G) in the GI head (input size 256×256). Results are reported as mean±std over three runs.

Groups	Params (M)	GFLOPs	Youden ↑	HTER (%) ↓
16	3.476	0.248	0.977 ± 0.004	1.16 ± 0.19
30	3.497	0.249	0.977 ± 0.010	1.14 ± 0.50
60	3.543	0.251	0.972 ± 0.020	1.41 ± 1.01
120	**3.635**	**0.256**	**0.984 ± 0.004**	**0.82 ± 0.21**
240	3.818	0.265	0.965 ± 0.020	1.77 ± 0.98

Table 13 Effect of GI block placement in MobileNetV3 (groups = 120). Results are reported as mean±std over three runs. GFLOPs are measured using a 256×256 input.

Placement	Params (M)	GFLOPs	Youden ↑	HTER (%) ↓
Beginning	2.975	0.257	0.968 ± 0.014	1.59 ± 0.70
End	3.635	0.256	0.984 ± 0.004	0.82 ± 0.21

Table 14 Performance for different input image sizes on the ROSE-Youtu dataset. Results are reported as mean±std over three runs.

Metric	**512 × 512**	**256 × 256**	**128 × 128**	**64 × 64**
Accuracy (%)	98.97 ± 0.43	98.90 ± 0.35	98.15 ± 1.01	93.48 ± 1.17
AUC-ROC	0.9994 ± 0.0003	0.9995 ± 0.0003	0.9971 ± 0.0033	0.9724 ± 0.0063
EER (%)	0.90 ± 0.36	0.82 ± 0.09	1.61 ± 0.80	8.21 ± 1.16
HTER (%)	0.61 ± 0.23	0.82 ± 0.21	1.64 ± 0.70	7.70 ± 1.42
Youden Index (YI)	0.988 ± 0.005	0.984 ± 0.004	0.967 ± 0.014	0.846 ± 0.028
Params (M)	3.635	3.635	3.635	3.635
GFLOPs	1.020	0.256	0.065	0.017

Moderate group sizes allow the model to capture sufficient spatial variation without unnecessary overhead. Excessive grouping (G=240) increases FLOPs and parameters without measurable benefit, while very low values reduce the network's representational capacity.

Effect of proposed GI placement

To assess how the position of the GI module affects performance, two placements were evaluated: (1) at the *beginning* of the MobileNetV3 backbone and (2) at the *end*, right before adaptive average pooling. Results are summarized in Table 13.

Placing the GI at the end yields superior results, indicating that context-adaptive filtering benefits most from high-level semantic representations. Early placement uses fewer parameters, but the two configurations have nearly identical GFLOPs.

Effect of input image size

To study the trade-off between input resolution and accuracy, models were trained with image sizes of 64×64, 128×128, 256×256, and 512×512. Results are reported in Table 14 and illustrated in Fig. 5. The t-SNE plots show a clear resolution-separation trend: at 64×64 the bona fide and spoof clusters overlap notably; at 128×128 and 256×256 clusters become more compact with wider margins; at 512×512 separation is strongest, matching the best EER/HTER.

Performance improves with higher resolutions, with the best results achieved at 512×512 (HTER 0.61%). However, computational cost grows rapidly, from 0.017 to 1.02 GFLOPs. The 256×256 resolution provides a balanced trade-off, offering near-maximum accuracy with manageable complexity, making it the most practical choice for real-time applications.

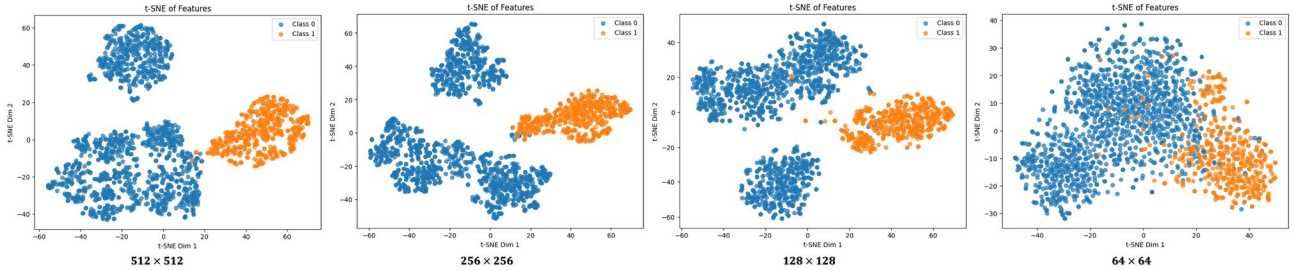


Fig. 5 t-SNE embeddings of features before the classification layer across different input resolutions. Higher resolutions yield better class separation.

Table 15 Effect of the reduction ratio (*reduce*$\in \{1, 4, 8\}$) in the end-placed involution[23] head ($G = 120$). Results are reported as mean±std over three runs.

Reduce	Params (M)	GFLOPs	Youden ↑	HTER (%) ↓
1	5.775	0.366	0.965 ± 0.028	1.75 ± 1.40
4	3.635	0.256	0.984 ± 0.004	0.82 ± 0.21
8	3.303	0.239	0.967 ± 0.016	1.65 ± 0.78

Table 16 Effect of the involution kernel size ($k \in \{3, 5, 7\}$) on the ROSE-Youtu dataset. Results are reported as mean±std over three runs.

Kernel Size	Params (M)	GFLOPs	EER (%) ↓	HTER (%) ↓
3×3	3.517	0.250	0.97 ± 0.28	0.99 ± 0.31
5×5	**3.635**	**0.256**	0.82 ± 0.09	0.82 ± 0.21
7×7	3.810	0.264	0.87 ± 0.18	0.92 ± 0.24

Effect of reduction ratio

Table 15 examines the influence of the reduction ratio (*reduce*) in the kernel generator of the involution head. As the bottleneck ratio increases, parameter count and FLOPs decrease, but performance follows a non-linear trend. The optimal configuration is *reduce*= 4, which delivers the highest Youden Index (0.984 ± 0.004) and lowest HTER (0.82 ± 0.21).

The results follow a non-monotonic trend across the evaluated reduction ratios. The setting *reduce*= 4 achieves the highest Youden index and lowest HTER while maintaining moderate computational cost, and is therefore selected as the default configuration.

Effect of kernel size

Table 16 examines the influence of the involution kernel size k on performance and computational complexity. The kernel size determines the spatial neighborhood used by the content-adaptive GI operator. All other settings, including the MobileNetV3-Large backbone, end-stage GI placement, reduction ratio $r = 4$, group count $G = 120$, input resolution, preprocessing, and training configuration, were kept fixed. Among the evaluated settings, $k = 5$ achieves the lowest EER and HTER, both equal to 0.82%.

The 3×3 kernel obtains an EER of 0.97% and an HTER of 0.99%. Increasing the kernel size to 5×5 improves both EER and HTER to 0.82%, while increasing the parameter count from 3.517M to 3.635M and the computational cost from 0.250 to 0.256 GFLOPs. A further increase to 7×7 raises the model size to 3.810M parameters and 0.264 GFLOPs, while slightly degrading performance to an EER of 0.87% and an HTER of 0.92%. These results indicate that expanding the spatial neighborhood beyond 5×5 does not provide additional performance gains and may incorporate less relevant context. Therefore, $k = 5$ is selected as the default kernel size because it provides the

best balance among local-detail preservation, contextual modeling, detection performance, and computational complexity.

Computational complexity and efficiency analysis

To examine computational efficiency, CASO-PAD is evaluated against representative lightweight architectures, including ShuffleNetV2[102], Once-for-All[105], MobileNetV2[103], MobileViTV2[106], GhostNet[107], and MobileNetV3[22]. All networks are trained and tested using identical settings with a 224×224 input resolution, ImageNet initialization, and consistent optimization parameters. Performance and complexity measurements are obtained under uniform benchmarking conditions to ensure fair comparison.

Figure 6 illustrates the relationship between spoof detection performance and model complexity from two complementary perspectives. The parameter-based analysis (left) shows that CASO-PAD attains the lowest HTER (1.02%) despite having only moderate model size (3.635M parameters), indicating superior parameter efficiency. Several smaller networks, such as ShuffleNetV2, exhibit noticeably higher error rates, suggesting that reduced parameter count alone does not guarantee robust liveness representation.

The GFLOPs-based view (right) further reveals that CASO-PAD achieves strong discriminative performance at low computational cost (0.19 GFLOPs). Importantly, models with comparable or even lower FLOPs do not consistently match this accuracy level, highlighting that effective architectural design and feature modeling play a more critical role than raw operation count. These observations collectively indicate that CASO-PAD improves generalization efficiency by extracting more informative representations per unit of computation rather than relying on larger networks.

Edge deployment: Jetson Orin Nano

Table 17 reports inference latency on the NVIDIA Jetson Orin Nano, reflecting realistic edge deployment conditions. While certain models exhibit slightly lower latency, these networks are associated with substantially higher HTER values (Fig. 6), indicating weaker spoof discrimination capability. CASO-PAD maintains competitive runtime performance while delivering significantly improved detection accuracy, demonstrating that modest increases in latency can be justified by notable gains in reliability and robustness.

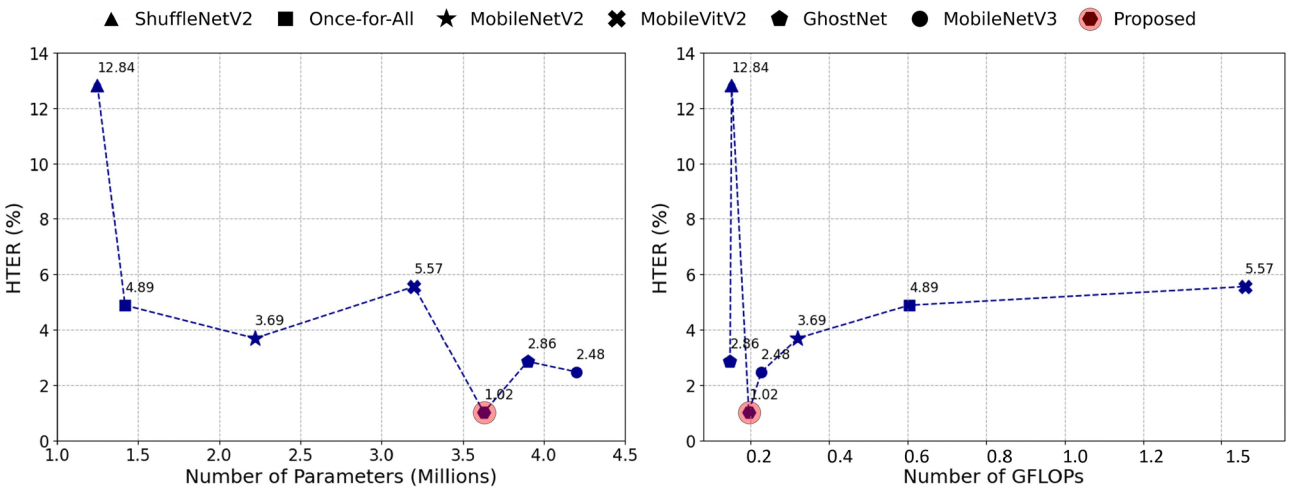


Fig. 6 Trade-off comparison between model complexity and performance. (Left) HTER vs. number of parameters. (Right) HTER vs. GFLOPs.

Table 17 Inference latency comparison on the NVIDIA Jetson Orin Nano for an input resolution of 224×224 using CUDA execution.

Method	Jetson Orin Nano Latency (ms)
ShuffleNetV2[102]	22.6 ± 0.6
Once-for-All[105]	49.36 ± 0.4
MobileViTV2[106]	28.87 ± 0.8
MobileNetV2[103]	19.0 ± 0.4
MobileNetV3 (large)[22]	23.17 ± 0.7
GhostNet[107]	37.8 ± 0.2
CASO-PAD (Proposed)	$\mathbf{25.6 \pm 0.8}$

Values denote average inference latency (mean ± std) over 100 runs on the NVIDIA Jetson Orin Nano in 25 W mode using CUDA.

These measurements support the practical feasibility of CASO-PAD on the evaluated NVIDIA Jetson Orin Nano configuration. Performance and latency on other mobile, embedded, and production platforms remain to be evaluated.

Quantitative region-wise grad-CAM analysis

Grad-CAM[108] was calculated at the output of the grouped involution head. For sample , let $G_i(x,y)$ denote the Grad-CAM response at spatial location (x,y). Its positive component was defined as

$$\widetilde{G}_i(x,y) = \max(0, G_i(x,y)). \tag{13}$$

The percentage of positive attribution assigned to region r was then calculated as

$$A_{i,r} = 100 \frac{\sum_{(x,y)\in r} \widetilde{G}_i(x,y)}{\sum_{x,y} \widetilde{G}_i(x,y)}. \tag{14}$$

Facial landmarks were used to divide each input into five mutually exclusive and collectively exhaustive regions: eyes, nose, mouth, remaining facial region, and background. Consequently, the regional attribution values sum to 100% for every analysed sample.

As shown in Fig. 7(a), presentation-attack decisions assigned the largest proportion of positive Grad-CAM attribution to the remaining facial region ($48.73 \pm 8.58\%$), followed by the background ($39.63 \pm 10.43\%$). The nose received $6.34 \pm 1.59\%$, whereas the mouth and eye regions accounted for $2.73 \pm 1.33\%$ and $2.56 \pm 0.79\%$, respectively. The combined attribution within the detected facial area was therefore $60.37 \pm 10.43\%$. These results indicate that presentation-attack decisions predominantly depend on spatially distributed evidence across the broader facial area rather than on one isolated internal facial component.

In contrast, bona fide decisions were dominated by background attribution ($83.43 \pm 8.17\%$). Within the detected face, the remaining facial region contributed $13.42 \pm 6.49\%$, followed by the nose ($1.63 \pm 1.04\%$), mouth ($0.95 \pm 0.66\%$), and eyes ($0.58 \pm 0.36\%$). The total facial attribution for bona fide samples was $16.57 \pm 8.17\%$, substantially lower than the corresponding $60.37 \pm 10.43\%$ observed for presentation attacks.

The resulting class-dependent distributions suggest that CASO-PAD uses different spatial evidence when forming bona fide and presentation-attack decisions. Specifically, attack predictions are more strongly associated with evidence within the detected facial area, whereas bona fide predictions are more strongly associated with contextual or background regions. The schematic projections in Fig. 7(b,c) summarize these class-level regional differences.

These findings should be interpreted as post-hoc attribution rather than causal evidence that a particular region determines the prediction. Furthermore, because the regions occupy different numbers of pixels, the reported

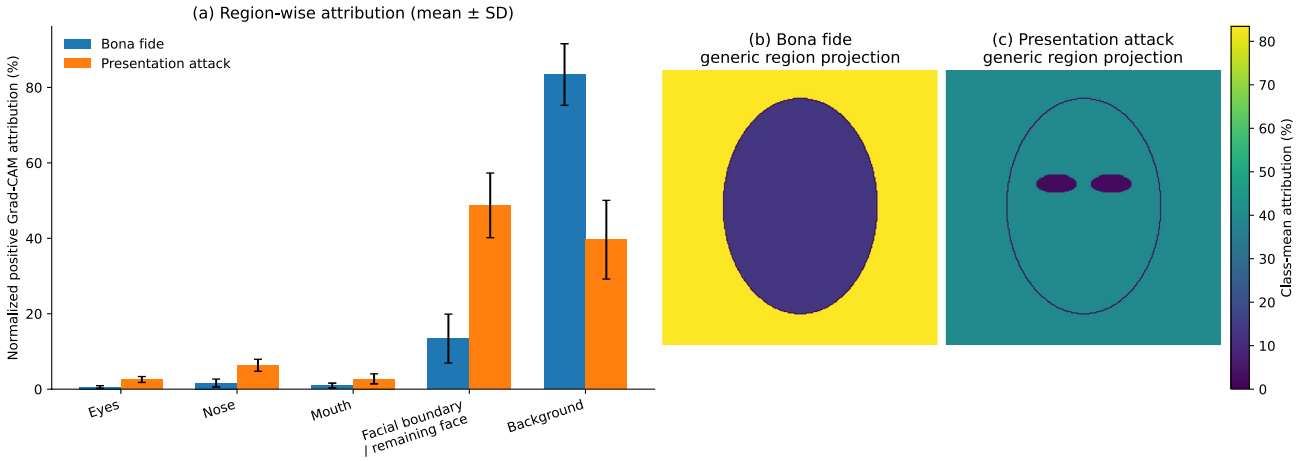


Fig. 7 Quantitative region-wise Grad-CAM analysis on the ROSE-Youtu test set. (a) Distribution of normalized positive Grad-CAM attribution across the eyes, nose, mouth, remaining facial region, and background for bona fide and presentation-attack decisions. Bars represent the class-wise mean and error bars denote one standard deviation across test samples. (b,c) schematic projections of the corresponding regional means for bonafide and presentation-attack samples, respectively. The schematics are region-level representations and are not pixel-wise averaged facial heatmaps..

percentages quantify the distribution of total positive Grad-CAM mass and not attribution density per unit area. In particular, the relatively high background attribution may be influenced partly by the larger spatial extent of the background region.

Limitations and future work

Despite its strong performance and computational efficiency, CASO-PAD has several limitations. First, the model has not been evaluated against digitally generated adversarial examples or adaptive physical attacks. Consequently, the results reported in this study should not be interpreted as evidence of adversarial robustness. Small, deliberately optimized perturbations may alter the content-dependent kernels generated by the grouped involution operator and potentially affect the final prediction.

Second, CASO-PAD operates on a single RGB frame and therefore does not exploit temporal motion, physiological, depth, acoustic, or other complementary liveness cues. This design supports lightweight deployment but may limit performance when spoofing evidence is weak in an individual frame or is primarily expressed through temporal behavior. Third, although the experiments include OULU-NPU Protocols I and II and four leave-one-dataset-out cross-domain evaluations, performance varies across unseen target domains. Therefore, generalization to arbitrary cameras, environments, presentation instruments, and attack distributions cannot be assumed. Moreover, the current experiments do not provide exhaustive evaluation against entirely unseen semantic attack categories.

Future work will therefore evaluate CASO-PAD against standard white-box and black-box adversarial attacks, including gradient-based and transfer-based perturbations, and investigate adversarial training, input denoising, and consistency regularization for improving robustness. Further evaluation will include additional unseen digital and physical attack categories and more diverse acquisition domains. Another direction is to incorporate lightweight temporal aggregation or complementary depth, physiological, or acoustic cues while preserving the model's current mobile-class computational efficiency.

Conclusion

This paper presented CASO-PAD, a lightweight, RGB-only, single-frame Face Presentation Attack Detection model that integrates a grouped involution head near the end of a MobileNetV3-Large backbone. The resulting architecture applies location- and group-dependent spatial filtering to semantically rich feature maps while retaining a compact, edge-oriented design.

Across the intra-dataset evaluations on Replay-Attack, Replay-Mobile, OULU-NPU, ROSE-Youtu, and SiW-Mv2 Protocol-1, CASO-PAD achieved strong performance. The additional OULU-NPU Protocol I and II experiments assess generalization to unseen environmental conditions and presentation instruments, while the four leave-one-dataset-out evaluations provide a broader assessment across unseen target domains. The variation observed among the cross-domain protocols indicates that performance remains dependent on the target acquisition distribution and should not be interpreted as evidence of universal real-world robustness.

Controlled ablations show that the performance gains arise from the task-specific integration and configuration of the GI head. The kernel audit and Grad-CAM analysis further provide evidence that CASO-PAD emphasizes relevant spatial, textural, and boundary cues. Measurements on the NVIDIA Jetson Orin Nano support practical feasibility under the evaluated hardware configuration, although performance and latency on other mobile, embedded, and production platforms remain to be established.

The implementation of CASO-PAD is publicly available at: https://github.com/Shujaat123/CASO-PAD.

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Author contributions S.K. conceived the experiments, conducted the experiments, analysed the results and also wrote and reviewed the manuscript.

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Data availability The datasets analyzed during the current study are publicly available from their respective sources, including Replay-Attack, Replay-Mobile, OULU-NPU, ROSE-Youtu, and SiW-Mv2. Links to these datasets are provided in the corresponding references. The implementation code is publicly available at: https://github.com/Shujaat123/CASO-PAD. No new datasets were generated during the current study.

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Authors and Affiliations

Shujaat Khan[1,2]

✉ Shujaat Khan
shujaat.khan@kfupm.edu.sa

[1] Department of Computer Engineering, College of Computing and Mathematics, King Fahd University of Petroleum & Minerals, Dhahran 31261, Saudi Arabia

[2] SDAIA–KFUPM Joint Research Center for Artificial Intelligence, College of Computing and Mathematics, King Fahd University of Petroleum & Minerals, Dhahran 31261, Saudi Arabia